

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Practice Program

Date: 06/08/2026

Week No.: 5

Program S. No. 9

Title	Web Scraping and Data Collection			
Problem Statement	Scrape data from a webpage and store it in a structured format like CSV or JSON format			
Objectives	• To understand the concept of web scraping and data collection. • To retrieve webpage content using the Requests library. • To extract required information from HTML using BeautifulSoup. • To organize the scraped data using Pandas DataFrame. • To store the collected data in structured formats such as CSV and JSON.			
Dataset URL Link / File	https://quotes.toscrape.com			
Function Details	Sl. No.	Function	Package	Description of Function
	1	requests.get()	requests	Retrieves the HTML content of a webpage
	2	BeautifulSoup()	bs4	Parses the HTML content of the webpage
	3	find_all()	BeautifulSoup	Finds all matching HTML elements
	4	find()	BeautifulSoup	Finds the first matching HTML element
	5	DataFrame()	pandas	Converts collected data into tabular format
	6	to_csv()	pandas	Stores data in CSV format
	7	json.dump()	json	Stores Python data in JSON format
	8	json.load()	json	Reads data from a JSON file
Steps/ Process/ Procedure/	1. Import the required libraries: requests, BeautifulSoup, pandas and json. 2. Specify the URL of the webpage to be scraped.			

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Algorithm/ Flowchart/ Model	3. Send an HTTP GET request using requests.get(). 4. Parse the HTML content using BeautifulSoup. 5. Locate the required HTML elements containing quotes, authors and tags. 6. Extract the required data from the webpage. 7. Store the extracted data in a Python list. 8. Convert the collected data into a Pandas DataFrame. 9. Save the structured data in CSV format using to_csv(). 10. Save the same data in JSON format using json.dump(). 11. Display the collected data to verify the output.
Program	<pre>import requests from bs4 import BeautifulSoup import pandas as pd import json url = "https://quotes.toscrape.com/" response = requests.get(url) print("Status Code:", response.status_code) soup = BeautifulSoup(response.text, "html.parser") print("Website Title:", soup.title.text) quotes = [] for item in soup.find_all("div", class_="quote"): quote = item.find("span", class_="text").text author = item.find("small", class_="author").text tags = [tag.text for tag in item.find_all("a", class_="tag")] quotes.append({ "Quote": quote, "Author": author, "Tags": " ".join(tags) }) print("Number of quotes collected:", len(quotes))</pre>

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```
df = pd.DataFrame(quotes)

df

df.to_csv("quotes.csv", index=False)

print("Data saved successfully in CSV format.")

with open("quotes.json", "w", encoding="utf-8") as file:
    json.dump(quotes, file, indent=4, ensure_ascii=False)

print("Data saved successfully in JSON format.")

csv_data = pd.read_csv("quotes.csv")

csv_data

all_quotes = []

page = 1

while True:

    url = f"https://quotes.toscrape.com/page/{page}/"

    response = requests.get(url)
    soup = BeautifulSoup(response.text, "html.parser")

    items = soup.find_all("div", class_="quote")

    if not items:
        break

    for item in items:

        quote = item.find("span", class_="text").text
        author = item.find("small", class_="author").text

        tags = [
            tag.text
            for tag in item.find_all("a", class_="tag")
        ]

        all_quotes.append({
            "Quote": quote,
```

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	<pre>"Author": author, "Tags": ", ".join(tags) }) page += 1 print("Total quotes collected:", len(all_quotes)) df = pd.DataFrame(all_quotes) df df.to_csv("quotes_dataset.csv", index=False) with open("quotes_dataset.json", "w", encoding="utf-8") as file: json.dump(all_quotes, file, indent=4, ensure_ascii=False) print("CSV and JSON files created successfully.")</pre>
Output Snapshots With Explanation/ Observations	The required libraries requests, BeautifulSoup, pandas and json are imported for web scraping, HTML parsing, data organization and structured data storage. <pre>1 import requests 2 from bs4 import BeautifulSoup 3 import pandas as pd 4 import json 1 url = "https://quotes.toscrape.com/" 2 3 response = requests.get(url) 4 5 print("Status Code:", response.status_code)</pre> Status Code: 200 <i>Fig. 1.1: Importing Required Python Libraries</i>

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The webpage <https://quotes.toscrape.com/> is accessed using `requests.get()`. The status code 200 confirms that the webpage was successfully retrieved.

```
[7]: 1 quotes = []
      2
      3 for item in soup.find_all("div", class_="quote"):
      4
      5     quote = item.find("span", class_="text").text
      6     author = item.find("small", class_="author").text
      7
      8     tags = [
      9         tag.text
     10         for tag in item.find_all("a", class_="tag")
     11     ]
     12
     13 quotes.append({
     14     "Quote": quote,
     15     "Author": author,
     16     "Tags": ", ".join(tags)
     17 })
     18
     19 print("Number of quotes collected:", len(quotes))

Number of quotes collected: 10
```

Fig. 1.2: Accessing the Webpage

The HTML content of the webpage is parsed using BeautifulSoup. The webpage title confirms that the correct page has been successfully loaded.

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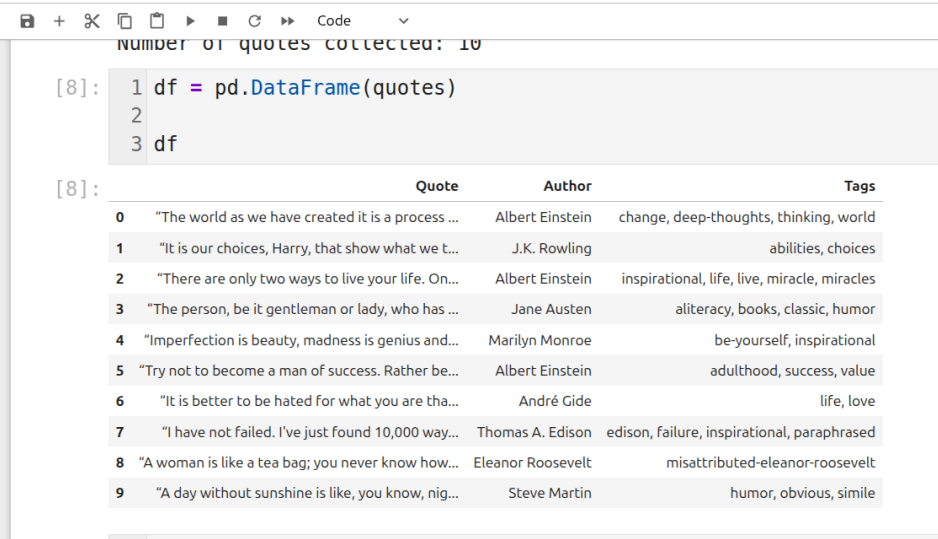
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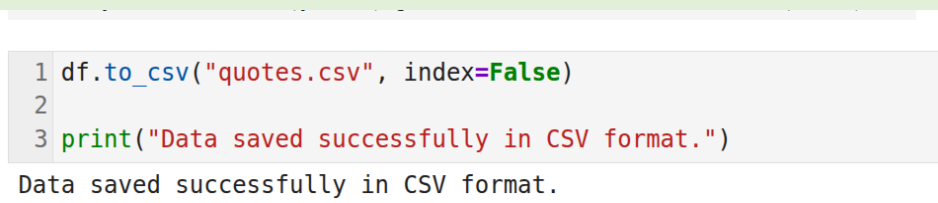
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```
[8]: 1 df = pd.DataFrame(quotes)
      2
      3 df
```

	Quote	Author	Tags
0	"The world as we have created it is a process ...	Albert Einstein	change, deep-thoughts, thinking, world
1	"It is our choices, Harry, that show what we t...	J.K. Rowling	abilities, choices
2	"There are only two ways to live your life. On...	Albert Einstein	inspirational, life, live, miracle, miracles
3	"The person, be it gentleman or lady, who has ...	Jane Austen	aliteracy, books, classic, humor
4	"Imperfection is beauty, madness is genius and...	Marilyn Monroe	be-yourself, inspirational
5	"Try not to become a man of success. Rather be...	Albert Einstein	adulthood, success, value
6	"It is better to be hated for what you are tha...	André Gide	life, love
7	"I have not failed. I've just found 10,000 way...	Thomas A. Edison	edison, failure, inspirational, paraphrased
8	"A woman is like a tea bag; you never know how...	Eleanor Roosevelt	misattributed-eleanor-roosevelt
9	"A day without sunshine is like, you know, nig...	Steve Martin	humor, obvious, simile

Fig. 1.3: Parsing the HTML and displaying the dataset Content



```
1 df.to_csv("quotes.csv", index=False)
2
3 print("Data saved successfully in CSV format.")
```

Data saved successfully in CSV format.

Fig. 1.4: Saving Data in CSV Format

The scraped data is successfully stored in CSV format using the pandas to_csv() function.

The scraped data is successfully stored in JSON format using the json.dump() function.

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```
1 with open("quotes.json", "w", encoding="utf-8") as file:
2     json.dump(quotes, file, indent=4, ensure_ascii=False)
3
4 print("Data saved successfully in JSON format.")
```

Data saved successfully in JSON format.

Fig. 1.5: Saving Data in JSON Format

	Quote	Author	Tags
0	"The world as we have created it is a process ...	Albert Einstein	change, deep-thoughts, thinking, world
1	"It is our choices, Harry, that show what we t...	J.K. Rowling	abilities, choices
2	"There are only two ways to live your life. On...	Albert Einstein	inspirational, life, live, miracle, miracles
3	"The person, be it gentleman or lady, who has ...	Jane Austen	aliteracy, books, classic, humor
4	"Imperfection is beauty, madness is genius and...	Marilyn Monroe	be-yourself, inspirational
5	"Try not to become a man of success. Rather be...	Albert Einstein	adulthood, success, value
6	"It is better to be hated for what you are tha...	André Gide	life, love
7	"I have not failed. I've just found 10,000 way...	Thomas A. Edison	edison, failure, inspirational, paraphrased
8	"A woman is like a tea bag; you never know how...	Eleanor Roosevelt	misattributed-eleanor-roosevelt
9	"A day without sunshine is like, you know, nig...	Steve Martin	humor, obvious, simile

Fig 1.6: Displaying the saved json and csv

The scraped data is successfully stored in JSON format using the `json.dump()` function.

The saved CSV file is read using `pandas.read_csv()` and displayed to verify that the data has been stored correctly.

The saved JSON file is read using `json.load()` and the stored records are displayed successfully, confirming correct JSON storage.

Conclusions

The experiment was successfully completed by scraping data from the given webpage using Python. The webpage content was retrieved using the requests library and parsed using BeautifulSoup. The required information such as quotes, authors, and tags was extracted and organized using a Pandas DataFrame. Finally, the collected data was successfully stored in structured CSV and JSON formats for further analysis and use.

References

1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018
2. Python Program (Google Colab link /Github link)
3. Class lecture link

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